

RAMCO INSTITUTE OF TECHNOLOGY

*DEPARTMENT
OF
ELECTRICAL AND ELECTRONICS ENGINEERING*

ELECTRO-TECHNOMANIA!



**ACADEMIC YEAR
2023-2024**

TECHNICAL MAGAZINE



About Ramco Institute of Technology



Ramco Institute of Technology is founded with a vision to impart high quality engineering education at an affordable cost. Under the able guidance of our Chairman Shri.P.R.Venketrama Raja, son of Shri P. R. Ramasubrahmaneya Rajha; distinguished professionals, academicians and education experts, we continually aim to revolutionize the learning environment by creating an enviable knowledge pool of engineering and technology graduates who are attuned to the current industry requirements.

Being part of the Ramco Group, which is well known for its qualitative and innovative brands not only in India but across the world, we are looked upon to set high standards in the education sector. True to its legacy, RIT has embarked on a mission to empower students with high-quality, accessible yet world-class engineering education and prepare these young minds for lifelong learning by creating and disseminating appropriate knowledge



Department of Electrical and Electronics Engineering

The Department of Electrical and Electronics Engineering is established in the year 2013. Currently, it offers a 4 year Under-Graduate programme, B.E. Electrical and Electronics Engineering with a sanctioned intake of 60.

The objective of the Department is to impart quality education and competitive training that will transform Students into Professional Engineers who are prepared to meet the needs of the society and adapt to the rapidly changing technology in various areas of Electrical and Electronics Engineering.

The Department has well-qualified and experienced Faculty members with specialization in Power Systems, High Voltage Engineering, Power Electronics and Control Systems. The Faculty members also have wide experience in teaching. They have a strong sense of responsibility to provide the finest possible engineering education to the students.

The Department has dedicated and state-of-the-art laboratories and leading softwares like DIgSILENT Lab VIEW Academy, ETAP, MATLAB, PLC/SCADA and Power World Simulator to cater to academic and research activities. It provides the Students with an in-depth knowledge in fundamentals and practical applications, so that he/she can pursue for wide-range of career opportunities.

The Department conducts Workshops, Seminars and Conferences and invites renowned academicians and industrial experts to enable the Students and Faculty to update with the latest developments. It also encourages Students to interact with Experts from Industries and other prominent Institutions.

Vision, Mission, PEOs & PSOs

Vision

- To create competent electrical and electronics engineers with professional and research skills, and a sense of ethics to meet the demands of multidisciplinary challenges and a technologically advancing society.

Mission

- Empowering the students with up-to-date knowledge and technological skills in Electrical and Electronics Engineering thrust areas.
- Providing a rigorous practical training through industrial collaboration to produce Electrical and Electronics Engineers with good design experience.
- Adopting effective teaching learning methodologies and value added courses to create Electrical and Electronics Engineers with a strong fundamental knowledge.
- Enabling the students to be life-long learners to meet intellectual, ethical and career challenges.

Program Educational Objectives (PEOs)

- Graduates will pursue their careers in diversified sectors of the industry, higher education in their fields of interest and/or emerge as entrepreneurs.
- Graduates will design, manufacture, and/or maintain systems in the field of electrical and electronics engineering and allied engineering industries.
- Graduates will work effectively as individuals and as team members in multidisciplinary projects with leadership qualities, social and ethical responsibilities.

Program Specific Outcomes (PSOs)

PSO1: Analyse the performance of electrical and electronic systems using modern software packages.

PSO2:Specify, design, prototype and test electronic systems and embedded systems in various engineering applications.

PSO3:Design and develop the hardware and software required for industrial automation systems.



INDEX



SENSORS IN SMART PHONE

110DC SUPPLY IN INDIAN TRAIN

BATTERY MANAGEMENT SYSTEM

**VEHICLE MONITORING BASED ON
LI-FI TECHNOLOGY**

**THE NEED FOR DIGITAL
MULTIMETERS**

**HYBRID ENERGIES AND MICROGRID
SYSTEM**

SENSORS

RASPBERRY PI IN AUTOMATION

FINGERPRINT SENSING PROCESS

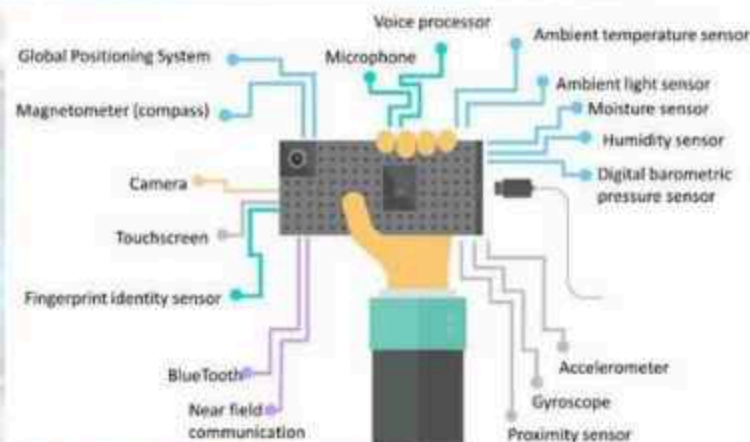
SENSORS IN SMART PHONE

In today's world, smartphones hold a significant place as versatile handheld devices that merge the features of a mobile phone with advanced computing capabilities.



Key Features

- **Touchscreen Interface:** The majority of smartphones are equipped with a capacitive touchscreen, allowing users to engage with the device through taps, swipes, and gestures.
- **Operating Systems:** The leading operating systems are Android and iOS, both providing a variety of apps, features, and user experiences.
- **Connectivity:** Smartphones offer multiple connectivity options, including:
 - Cellular Networks (2G, 3G, 4G, 5G)
 - Wi-Fi for internet access
 - Bluetooth for peripheral connections
 - NFC for contactless payments and data transfer
- **Cameras:** Most smartphones include multiple cameras (both front and rear), facilitating high-quality photography, video recording, and functionalities like portrait and night modes.
- **Apps:** Users have access to a vast selection of applications from app stores, encompassing everything from social media and productivity to gaming and fitness.



Advantages

- **Portability:** Smartphones are compact and lightweight, allowing for convenient use while on the move.
- **Multifunctionality:** They function as communication devices, cameras, navigation aids, gaming consoles, and much more.
- **Internet Access:** With smartphones, users enjoy seamless internet access for social media, browsing, and online shopping.
- **Personalization:** Users can tailor their devices with apps, wallpapers, and settings to reflect their individual preferences.

Sensors: As previously mentioned, smartphones are equipped with various sensors that enhance their capabilities, such as accelerometers, gyroscopes, proximity sensors, and more.

M SANDHIYA III YEAR EEE

110 DC SUPPLY IN INDIAN TRAIN

Indian Railways is one of the largest railway networks in the world, crucial to India's transportation. All trains use a standardized 110 V DC supply for applications like fans, lighting, and charging ports, designed for efficient operation.



There are several technical reasons follows



- Historical standardization facilitated easy sourcing of components with the introduction of electric trains.
- Equipment Compatibility: Many existing systems and components are designed for 110 V DC, avoiding costly upgrades.
- Safety: Lower voltage reduces the risk of electric shock.
- Battery Compatibility: Many auxiliary systems work efficiently with 110 V DC.
- Cost Efficiency: Familiarity with the standard helps reduce maintenance costs.
- Motor Performance: DC motors provide variable speed and torque, ideal for train operations.

N S PRAVEEN KUMAR - III YEAR EEE

The use of 110 V DC in Indian Railways is based on historical precedent, practical factors, and technical needs for train operations. Despite newer systems using different voltages, established infrastructure and component compatibility have kept 110 V DC in specific applications.

Battery management system

A Battery Management System is an electronic system designed to manage a rechargeable battery or battery pack, ensuring optimal performance, safety, and longevity. The BMS achieves this through a combination of monitoring, control, and communication functions.



KEY COMPONENTS:

- **Monitoring:** A BMS continuously monitors the voltage, current, and temperature of each cell in the battery pack. This data is crucial for assessing the overall health and state of charge (SoC) of the battery.
- **Balancing:** In a battery pack, individual cells may have slight variations in capacity and state of charge. A BMS balances these cells to ensure they charge and discharge evenly, which enhances performance and extends the battery's lifespan.
- **Protection:** A BMS protects the battery from conditions that could lead to damage or safety hazards, such as overcharging, deep discharging, and overheating. It can disconnect the battery or limit current flow when unsafe conditions are detected.
- **Communication:** Modern BMS units often include communication interfaces to relay data to external systems, such as vehicle control units or charging stations. This feature allows for real-time monitoring and management of the battery.
- **State of Health (SoH) Assessment:** The BMS evaluates the overall health of the battery, helping to predict its lifespan and performance over time.

Applications of BMS:

*Renewable Energy Storage
Grid Energy Storage
Consumer Electronics*

Trends in BMS Future:

*Wireless Monitoring and Control
Artificial Intelligence (AI) Integration
Advanced Materials and Battery Chemistries*

CONCLUSION:

As we look towards a future driven by electric mobility and energy storage solutions, investing in advanced BMS technology will be crucial for maximizing the potential of battery systems.

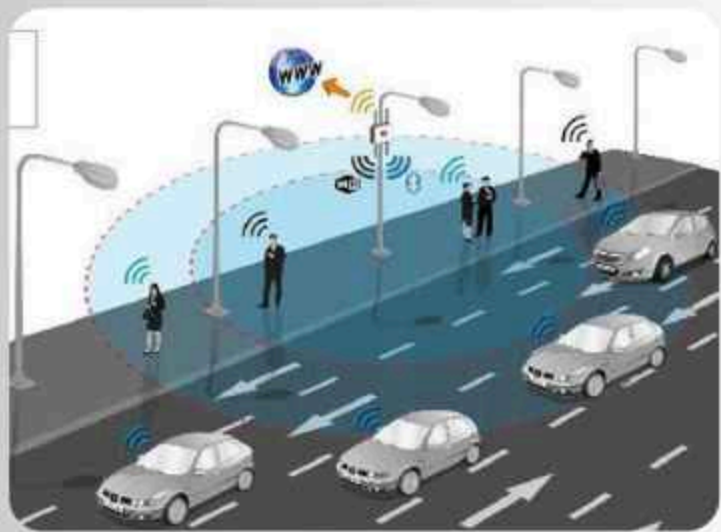
Ananth.S II year EEE

VEHICLE MONITORING BASED ON LIFI TECHNOLOGY



This work provides an intelligent system for accident prevention and detection for human life safety. That prevention part has various sensor like eye blink sensor, alcohol sensor and ultrasonic sensor.

Now a days, traffic accident detection is becoming one of the interesting fields due to its tremendous application in intelligent transportation systems. Main causes behind these road accidents are lack of unskilled drivers, consuming alcohol while driving, over speed, sleep while driving. Plenty of solutions have been applied to prevent these road accidents. But most of them were failed to prevent this. In this work we present an advanced accident detection using LIFI technology.



If the sensor detects whether the rider consumes alcohol or the distance between two vehicles is low then it sends that information to another vehicle which is going in front it. So that they can be alert. And if the driver is sleeping while driving means then the eye blink sensor detects it and give an alert to the driver.

BASKAR.G, II YEAR EEE



The Need for Digital Multimeters

KOPPERUNDEVIN III YEAR EEE

- Traditional analog multimeters struggle with inaccuracies due to parallax errors and limited resolution, complicating precise measurements and slowing down troubleshooting. This highlights the need for digital multimeters (DMMs), which provide accurate, real-time readings and enhanced functionalities, improving reliability in electrical diagnostics.
- Digital multimeters measure voltage, current, and resistance with clear displays for easy interpretation. The terms "counts" and "digits" refer to their measurement accuracy, with some detecting minute variations, such as 1 millivolt (1 mV). DMMs can measure AC and DC voltage, current flow, and resistance, and often include continuity and diode testing, making them essential for troubleshooting and DIY projects.

How to Use a Digital Multimeter

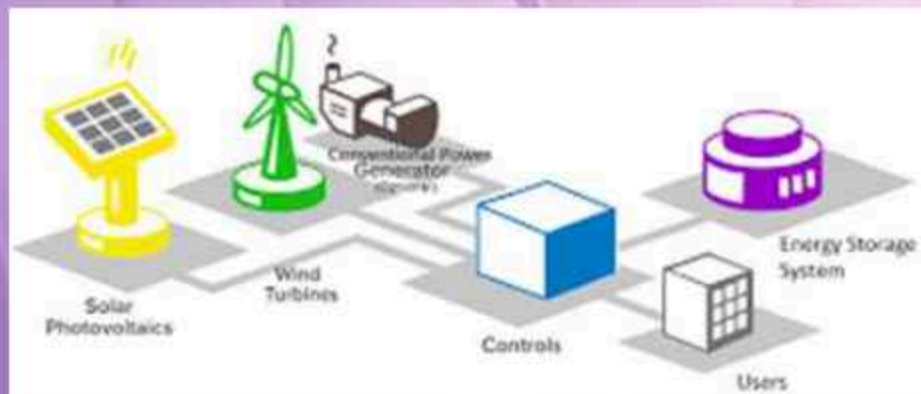
Using a DMM is straightforward: Multimeter

1. **Set Up:** Turn on the DMM and select the measurement mode (voltage, current, or resistance), choosing AC or DC as needed.
2. **Connect Probes:** Insert the black probe into the COM port and the red probe into the appropriate port for your measurement.
3. **Measure Voltage:** Connect the probes across the component or power source (black to negative, red to positive), then read the display.
4. **Measure Current:** Break the circuit to insert the multimeter in series, ensuring it's set to the correct current range.
5. **Measure Resistance:** Make sure the circuit is powered off, then connect the probes across the component and read the value.
6. **Test Continuity:** Set the DMM to continuity mode; a beep indicates a complete circuit.

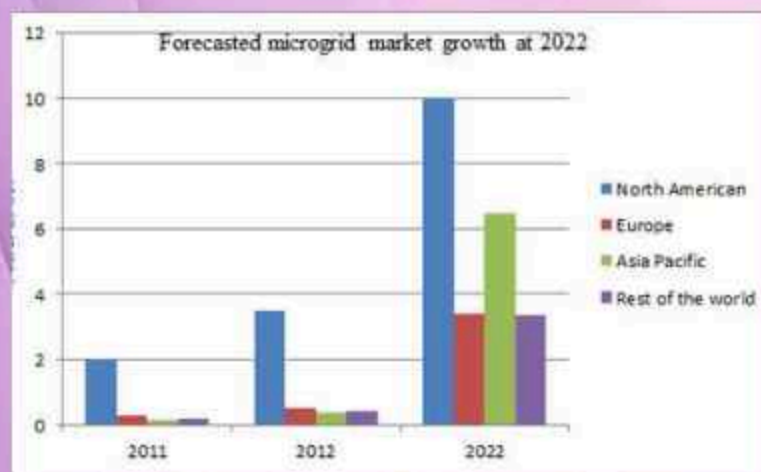


Hybrid energies and microgrid system

In today's world, the demand for energy is rising, driven by advancements in technology, with renewable energy playing a vital role in meeting this demand. A hybrid energy microgrid system is a self-sustaining energy solution that integrates various power sources to produce electricity. These systems can function independently or alongside the main power grid. Hybrid energy systems and microgrid networks represent a groundbreaking approach to energy management, utilizing solar, wind, and other fuels to enhance resilience, efficiency, and sustainability.



These self-sufficient networks can operate autonomously or be linked to the grid. Their key components include advanced control systems, energy storage systems, and smart grid technology, which oversee energy transfer and consumption.



The benefits are significant: the grid becomes more reliable and less prone to disruptions, fossil fuel usage is reduced, energy imports decrease, and environmental impacts from fuel combustion are minimized. Challenges such as high installation costs, regulatory hurdles, and the need for all systems to operate independently still exist.

M THIVYA III YEAR EEE

SENSORS



A **SENSOR** is a device that detects and responds to physical input from the environment. This input could be various forms of energy, such as light, heat, motion, moisture, pressure, or chemical composition. Sensors convert these physical signals into electrical signals, allowing other systems or devices to interpret the information and take appropriate action.

LIST OF SENSOR TYPES:

- Temperature Sensors
- Pressure Sensors
- Flow Sensors
- Smoke Sensors
- Ultrasonic Sensors
- Infrared (IR) Sensors
- Touch Sensors

FUTURE TRENDS:

1. **Miniaturization and Low Power Consumption**
2. Sensors are becoming smaller and more energy-efficient, enabling them to be embedded in tiny devices, wearables, and even implanted in the human body.
3. **Smart Sensors with Embedded AI**
4. Future sensors are expected to have onboard processing power and AI capabilities, allowing them to analyze data at the source. These "smart sensors" reduce the need for continuous data transmission, improving response times and data privacy.
5. **Multi-Sensor Integration**
6. Combining multiple sensors into a single device provides richer, more reliable data. For example, integrating temperature, humidity, and air quality sensors in smart home systems can provide a more comprehensive environmental analysis.
7. **Flexible and Wearable Sensors**
8. Wearable and flexible sensors are advancing rapidly, with applications in healthcare monitoring (e.g., heart rate, glucose levels) and smart textiles. These sensors are thin, flexible, and can be integrated into clothing or directly applied to the skin.



ADVANTAGES:

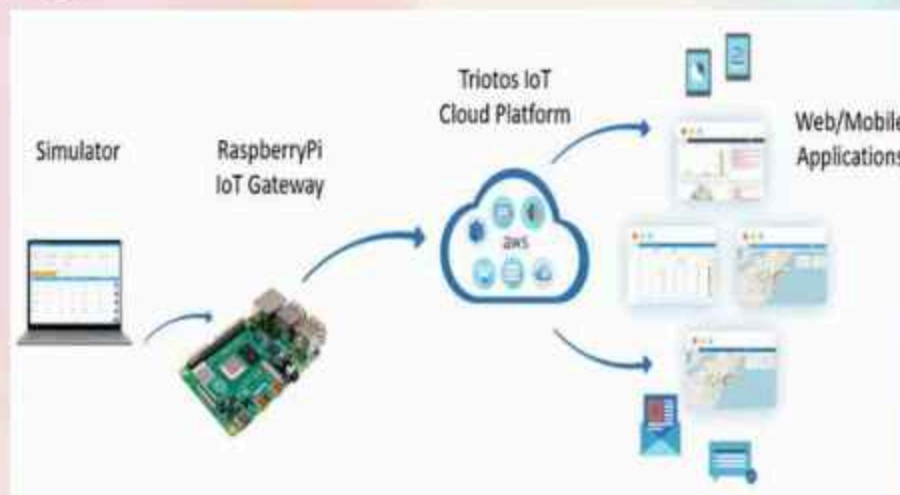
1. Real-Time Data Collection
2. Automation and Efficiency
3. Improved Safety
4. Enhanced Precision
5. Cost Savings
6. Better Decision-Making
7. Customization

DISADVANTAGES:

1. High Initial Cost
2. Power Consumption
3. Data Security and Privacy Risks
4. Signal Interference
5. Maintenance Needs
6. Limited Range and Coverage
7. Data Accuracy Variability

RASPBERRY PI - IN AUTOMATION

The Raspberry Pi is a small, affordable single-board computer created by the IT Foundation, perfect for intricate projects that require internet connectivity and advanced features. Operating at 1 GHz, it can control robots, track weather conditions, and manage smart home systems by linking sensors to the internet. With its integrated Ethernet and Wi-Fi, it facilitates real-time data communication, making it ideal for applications such as surveillance robots and weather monitoring systems. In summary, the Raspberry Pi is a versatile automation tool, beloved by both enthusiasts and professionals for its cost-effectiveness, compact size, and strong community support.



PROJECTS IN AUTOMATION

- Voice Controlled Home Automation Using Raspberry Pi
- Motion Based Time Lapse Camera With Optimized Storage
- Motion Activated Wildlife Recording Camera Raspberry Pi
- Drunk Driving Detection With Car Ignition Locking
- Smart Stand-up wheelchair using Raspberry Pi & RF
- IOT based Anti-theft Flooring System using Raspberry Pi



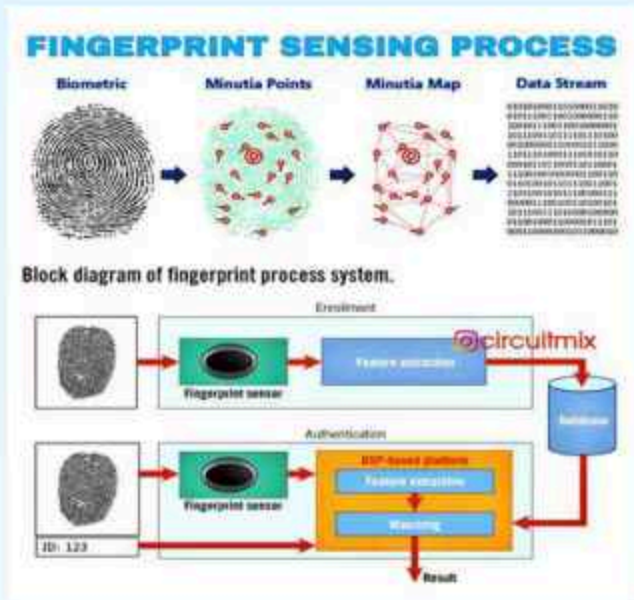
Choose Raspberry Pi if:

- You need a more powerful computing platform for complex tasks.
- You want to run a web server or handle multimedia.
- You're comfortable with a steeper learning curve.

-BHASKAR SETHUPATHI B
III YEAR EEE

FINGERPRINT SENSING PROCESS

From ATMs to the smartphone in your pocket, fingerprint scanners have become commonplace these days. Low price phones have a feature these days, sometimes alongside other biometric unlocking options like facial recognition. The technology has also moved on a lot from its early iterations, becoming faster and more accurate at capturing your fingerprint. Fingerprint scanners are a type of biometric security system which considered as good alternative to passwords and keys because fingerprints are unique and cannot be lost, lent, or left behind.



Sensing Technology

- Optical, capacitive, and ultrasonic sensors capture fingerprints using light, electrical changes, or sound waves.

Image Capture

- The sensor captures a high-quality fingerprint image when the user places their finger on it.

Image Processing

- The image is enhanced with noise reduction and contrast adjustments, then converted to binary.

Feature Extraction

- Key features like minutiae points and ridge patterns are extracted to create a digital template.

Template Storage

- The fingerprint template is securely stored in a compact format for efficient retrieval.

Fingerprint Matching

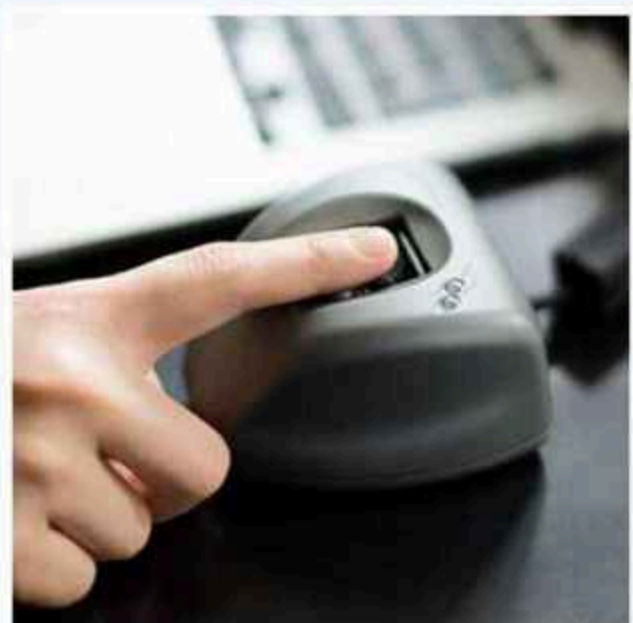
- The new fingerprint is compared to stored templates using matching algorithms.

Decision Making

- Access is granted if a match is found, or denied if no match exists.

Security Measures

- Templates are encrypted, and anti-spoofing technology ensures the fingerprint is real.





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